

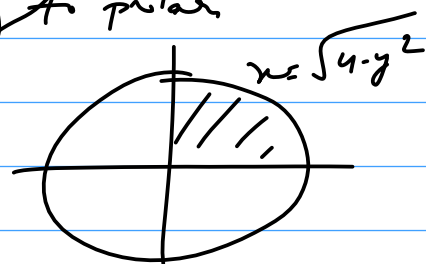
Calculus II: 2025 spring

1.a) $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2+y^2) dx dy$ by changing to polar

Solⁿ, Region: $0 \leq x \leq \sqrt{4-y^2}$
 $0 \leq y \leq 2$

Change to polar,

$0 \leq r \leq 2$, $0 \leq \theta \leq \pi/2$



$\Rightarrow \int_0^{\pi/2} \int_0^2 \cos(r^2) \cdot r dr d\theta$

Let $r^2 = u$

$du = 2r dr$

When $r=0$, $u=0$

$r=2$, $u=4$

$= \int_0^{\pi/2} \int_0^4 \cos u \cdot \frac{du}{2} d\theta$

$= \frac{1}{2} \int_0^{\pi/2} [\sin u]_0^4 d\theta \Rightarrow \frac{1}{2} \int_0^{\pi/2} \sin 4 d\theta$

$= \frac{\pi}{4} \sin 4 //$

2.a) Laplace of $\frac{\sin^2 t}{t}$
 i)

We know, $L\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s) ds$

$= \int_s^\infty L(\sin^2 t) ds \Rightarrow \int_s^\infty L\left(\frac{1 - \cos 2t}{2}\right) ds$

$= \frac{1}{2} \int_s^\infty L(1 - \cos 2t) ds = \frac{1}{2} \int_s^\infty \left(\frac{1}{s} - \frac{s}{s^2+2^2}\right) ds$

$= \frac{1}{2} \left[\log s - \frac{1}{2} \log(s^2+4) \right]_s^\infty$

$= \frac{1}{4} \left[2 \log s - \log(s^2+4) \right]_s^\infty \Rightarrow \frac{1}{4} \left[\log s^2 - \log(s^2+4) \right]_s^\infty$

$= \frac{1}{4} \left[\log\left(\frac{s^2}{s^2+4}\right) \right]_s^\infty \Rightarrow \frac{1}{4} \left[\log\left(\frac{1}{1+\frac{4}{s^2}}\right) \right]_s^\infty$

$\Rightarrow \frac{1}{4} \left[\log\left(\frac{1}{1+\frac{4}{s^2}}\right) - \log\left(\frac{s^2}{s^2+4}\right) \right]$

$= \frac{1}{4} \log\left(\frac{s^2+4}{s^2}\right) //$

$$\therefore f(t) = L^{-1} \left(\frac{w}{s^2(s^2+w^2)} \right)$$

$$\frac{w}{s^2(s^2+w^2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+w^2}$$

$$w = As(s^2+w^2) + B(s^2+w^2) + (Cs+D)s^2$$

$$w = \underline{A}s^3 + \underline{A}w^2s + \underline{B}s^2 + \underline{B}w^2 + \underline{C}s^3 + \underline{D}s^2$$

$$w = s^3(A+C) + s^2(B+D) + s(Aw^2) + (Bw^2)$$

Comparing like terms,

$$A+C=0, \quad B+D=0, \quad Aw^2=0, \quad Bw^2=w$$

$$A=-C, \quad B=-D, \quad \therefore A=0, \quad \therefore B=\frac{1}{w}$$

$$\therefore C=0, \quad \therefore D=-\frac{1}{w}$$

$$\therefore f(t) = L^{-1} \left[\frac{1}{ws^2} - \frac{1}{w(s^2+w^2)} \right]$$

$$= \frac{t}{w} - \frac{\sin wt}{w^2}$$

$$= \frac{wt - \sin wt}{w^2}$$

$$3.a) y'' + y = 0 \rightarrow (1)$$

Let $y = \sum_{n=0}^{\infty} a_n x^n \rightarrow (2)$ be solⁿ of (1),

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Put y', y in (1).

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0$$

Put $k=n-2$ and replace k with n for 1st series,

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\text{or, } (n+1)(n+2) a_{n+2} = -a_n$$

$$\therefore a_{n+2} = \frac{-a_n}{(n+1)(n+2)}$$

$$(n+1)(n+2)$$

For $n=0$, $a_2 = \frac{-a_0}{2}$	For $n=1$, $a_3 = \frac{-a_1}{6}$	For $n=2$, $a_4 = \frac{-a_2}{12}$ $= \frac{a_0}{24}$	For $n=3$, $a_5 = \frac{-a_3}{20}$ $= \frac{a_1}{120}$
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Put coefficients in (2),

$$y = a_0 + a_1 x - \frac{a_0 x^2}{2} - \frac{a_1 x^3}{6} + \frac{a_0 x^4}{24} + \frac{a_1 x^5}{120} - \dots$$

$$y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)$$

$$\therefore y = a_0 (\cos x) + a_1 (\sin x) //$$

3. b) Rodriguez formula:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \left[(x^2 - 1)^n \right]$$

$$\text{let } y = (x^2 - 1)^n$$

$$y_1 = n(x^2 - 1)^{n-1} \cdot 2x$$

$$(x^2 - 1)y_1 = 2nx(x^2 - 1)^n$$

$$(x^2 - 1)y_1 = 2nxy$$

$$(1 - x^2)y_1 + 2nxy = 0$$

Differentiate (n+1) times using Leibnitz theorem

$$(1 - x^2)y_{n+2} + {}^{n+1}C_1(-2x)y_{n+1} + {}^{n+1}C_2(-2)y_n + 2n[xy_{n+1} + {}^nC_1 y_n] = 0$$

$$\text{or, } (1 - x^2)y_{n+2} + (n+1)(-2x)y_{n+1} + \frac{(n+1)!}{(n-1)!2!}(-2)y_n + 2nxy_{n+1} + (n+1)2ny_n = 0$$

$$\text{or, } (1 - x^2)y_{n+2} + (y_{n+1})(-2x(n+1) + 2nx) + y_n((n+1)2n - n(n+1)) = 0$$

$$\text{or, } (1 - x^2) \frac{d^2(y_n)}{dx^2} - 2x \frac{d(y_n)}{dx} + n(n+1)y_n = 0$$

which is Legendre's differential eqn in y_n has it's polynomial solⁿ given by

$$P_n(x) = C y_n$$

$$= C \frac{d^n}{dx^n} \left[(x^2 - 1)^n \right]$$

$$= C \frac{d^n}{dx^n} \left[(x+1)^n (x-1)^n \right]$$

$$= C \left[(x-1)^n \frac{d^n}{dx^n} (x-1)^n + {}^nC_1 n(x-1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} (x+1)^n + \dots + (x+1)^n \frac{d^n}{dx^n} (x-1)^n \right]$$

$$= C \left[(x-1)^n n! + {}^nC_1 n(x-1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} (x+1)^n + \dots + (x+1)^n n! \right]$$

$$\text{Put } x=1, \quad P_n(1) = c [0 + 1 + \dots + 2^n \cdot 2!]$$

$$\text{or, } 1 = c(2^n 2!)$$

$$\therefore c = \frac{1}{2^n n!}$$

Substituting (in $P_n(x) = c y_n$

$$\therefore P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \left[(x^2-1)^n \right]$$

4.a) $D_n(\vec{f}) = ?$, $\vec{f} = xy^2 + yz^3$ at $P(2, -1, 1)$ along direction of

Solⁿ,
 $\nabla \cdot \vec{f} = \vec{i}(y^2) + \vec{j}(2xy + z^3) + \vec{k}(3yz^2)$ normal to surface $x \log z - y^2 + 4 = 0$
 at $P(2, -1, 1)$ at $(-1, 2, 1)$

$$\nabla \cdot \vec{f} = \vec{i} - 3\vec{j} - 3\vec{k}$$

$$\text{Let } \vec{a} = x \log z - y^2 + 4$$

The normal to the surface is given by,

$$\nabla \vec{a} = \vec{i}(\log z) + \vec{j}(-2y) + \vec{k}\left(\frac{x}{z}\right)$$

$$= \log z \vec{i} - 2y \vec{j} + \frac{x}{z} \vec{k}$$

$$\text{at } (-1, 2, 1); \quad \nabla \vec{a} = 0 \vec{i} - 4 \vec{j} - \vec{k}$$

Then,

$$D_n(\vec{f}) = \frac{(\nabla \cdot \vec{f}) \cdot \nabla \vec{a}}{|\nabla \vec{a}|} = \frac{(1, -3, -3) \cdot (0, -4, -1)}{\sqrt{4^2 + 1^2}} = \frac{15}{\sqrt{17}}$$

4.b) $\phi = \log(x^2 + y^2 + z^2)$, $\nabla \cdot (\text{grad } \phi) = ?$, $\nabla \times (\text{grad } \phi) = ?$

Solⁿ,

$$(\text{grad } \phi) = \left(\frac{2x}{x^2 + y^2 + z^2} \right) \vec{i} + \left(\frac{2y}{x^2 + y^2 + z^2} \right) \vec{j} + \left(\frac{2z}{x^2 + y^2 + z^2} \right) \vec{k}$$

$$\nabla \cdot (\text{grad } \phi) = \frac{2(x^2 + y^2 + z^2) - 4x^2}{(x^2 + y^2 + z^2)^2} + \frac{2(x^2 + y^2 + z^2) - 4y^2}{(x^2 + y^2 + z^2)^2} + \frac{2(x^2 + y^2 + z^2) - 4z^2}{(x^2 + y^2 + z^2)^2}$$

$$= 2 \left(\frac{y^2 + z^2 - x^2 + x^2 + z^2 - y^2 + x^2 + y^2 - z^2}{(x^2 + y^2 + z^2)^2} \right)$$

$$= \frac{2}{x^2 + y^2 + z^2}$$

$$\nabla \times (\text{grad } \phi) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{2x}{x^2+y^2+z^2} & \frac{2y}{x^2+y^2+z^2} & \frac{2z}{x^2+y^2+z^2} \end{vmatrix}$$

$$= \vec{i} \left(\frac{4yz}{(x^2+y^2+z^2)^2} - \frac{4yz}{(x^2+y^2+z^2)^2} \right) - \vec{j} \left(\dots \right)$$

$$= \underline{\underline{0}}$$

4. (c) Show integral is exact & evaluate

$$\int_{(0,2,3)}^{(1,1,1)} [yz \sinh(xz) dx + \cosh(xz) dy + xy \sinh(xz) dz]$$

(0,2,3)

Solⁿ,

$F_1 = yz \sinh(xz)$	$F_2 = \cosh(xz)$	$F_3 = xy \sinh(xz)$
$\frac{\partial F_1}{\partial y} = z \sinh(xz)$	$\frac{\partial F_2}{\partial x} = z \sinh(xz)$	$\frac{\partial F_3}{\partial x} = xyz \cosh(xz) + y \sinh(xz)$
$\frac{\partial F_1}{\partial z} = xyz \cosh(xz) + y \sinh(xz)$	$\frac{\partial F_2}{\partial z} = x \sinh(xz)$	$\frac{\partial F_3}{\partial y} = x \sinh(xz)$

Since, $\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$, $\frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}$, $\frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}$

$\therefore$ The integral F is exact.

Its solⁿ is given by

$$\Rightarrow \int F_1 dx + \int (\text{terms of } F_2 \text{ free from } x) dy + \int (\text{terms of } F_3 \text{ free from } x \text{ and } y) dz$$

$$\Rightarrow \int_{(0,2,3)}^{(1,1,1)} (yz \sinh xz dx + 0 + 0)$$

(0,2,3)

$$\Rightarrow yz \left[\frac{\cosh xz}{z} \right]_{(0,2,3)}^{(1,1,1)}$$

$$\Rightarrow [1 \cdot \cosh 1 - 2 \cdot \cosh 0]$$

$$\Rightarrow \cosh 1 - 2 //$$

5. a) $\iint_S \vec{F} \cdot \vec{N} ds = ?$ for hemisphere, $z = \sqrt{9-x^2-y^2}$

$$\vec{F} = 3x\vec{i} + 3y\vec{j} + z\vec{k}$$

$$= 3x\vec{i} + 3y\vec{j} + (\sqrt{9-x^2-y^2})\vec{k}$$

Let $\vec{r}$ be position vector

$$\vec{r} = x\vec{i} + y\vec{j} + (\sqrt{9-x^2-y^2})\vec{k}$$

$$\vec{r}_x = \vec{i} + 0\vec{j} - \left(\frac{x}{\sqrt{9-x^2-y^2}}\right)\vec{k}, \quad \vec{r}_y = 0\vec{i} + \vec{j} - \left(\frac{y}{\sqrt{9-x^2-y^2}}\right)\vec{k}$$

$$\vec{N} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{-x}{\sqrt{9-x^2-y^2}} \\ 0 & 1 & \frac{-y}{\sqrt{9-x^2-y^2}} \end{vmatrix} = \vec{i} \left(0 + \frac{x}{\sqrt{9-x^2-y^2}} \right) - \vec{j} \left(\frac{-y}{\sqrt{9-x^2-y^2}} \right) + \vec{k}$$

$$= \left(\frac{x}{\sqrt{9-x^2-y^2}} \right) \vec{i} + \left(\frac{y}{\sqrt{9-x^2-y^2}} \right) \vec{j} + \vec{k}$$

$$\vec{F} \cdot \vec{N} = \frac{3x^2}{\sqrt{9-x^2-y^2}} + \frac{3y^2}{\sqrt{9-x^2-y^2}} + \sqrt{9-x^2-y^2}$$

$$= \frac{3x^2 + 3y^2 + 9 - x^2 - y^2}{\sqrt{9-x^2-y^2}}$$

$$= \frac{2x^2 + 2y^2 + 9}{\sqrt{9-x^2-y^2}}$$

Taking projection on xy plane $z=0$

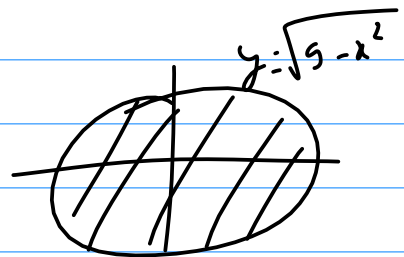
$$x^2 + y^2 = 9$$

$$y = \pm \sqrt{9-x^2}$$

Take $y=0$, $x = \pm 3$

change to polar,

$$0 \leq r \leq 3 \quad 0 \leq \theta \leq 2\pi$$



$$= \int_0^{2\pi} \int_0^3 \frac{(2r^2 + 9) r dr d\theta}{\sqrt{9-r^2}} \Rightarrow \int_0^{2\pi} \frac{(2r^3 + 9r)}{\sqrt{9-r^2}} dr d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \frac{(54 \sin^3 \phi + 27 \sin \phi) \cdot 3 \cos \phi d\phi d\theta}{3 \cos \phi} \quad \text{Put } r = 3 \sin \phi$$

$$= \int_0^{2\pi} \left[\frac{54}{2} \times \frac{\frac{1}{2}}{\frac{\sqrt{5}}{2}} + \frac{27}{2} \times \frac{1}{\frac{\sqrt{2}}{2}} \right] d\theta \quad \text{When, } r=0, \phi=0$$

$$r=3, \phi=\pi/2$$

$$= \int_0^{2\pi} \left[\frac{27 \times 2}{3} \times \frac{2}{1} \times \frac{1}{\sqrt{\pi}} + \frac{27}{2} \times \frac{2}{1} \times \frac{1}{\sqrt{\pi}} \right] d\theta$$

$$= \int_0^{2\pi} (36 + 27) d\theta$$

$$= 63 \times 2\pi$$

$$= 126\pi //$$

5.b) Gauss

$$\vec{F} = y^3 \vec{i} + x^3 \vec{j} + z^3 \vec{k}$$

$$S: x^2 + 4y^2 = 1, 0 < z < 5, x > 0, y > 0$$

$$\nabla \cdot \vec{F} = 3z^2$$

$$\text{Limit, } y = \frac{1}{2} \sqrt{1-x^2}, \text{ put } y=0, x=1$$

$$\iiint_S \vec{F} \cdot \vec{n} dA = \int_0^5 \int_0^1 \int_0^{\frac{1}{2}\sqrt{1-x^2}} (\nabla \cdot \vec{F}) dV = \iiint 3z^2 dy dx dz$$

$$\Rightarrow \int_0^5 \int_0^1 \left(3z^2 \times \frac{1}{2} \sqrt{1-x^2} \right) dx dz$$

$$= \frac{3}{2} \int_0^5 \int_0^1 z^2 (\sqrt{1-x^2}) dx dz$$

$$= \frac{3}{2} \int_0^5 z^2 \int_0^{\pi/2} \cos \theta \cdot \cos \theta d\theta dz$$

$$\text{let } x = \sin \theta$$

$$dx = \cos \theta d\theta$$

$$\text{when } x=0, \theta=0$$

$$x=1, \theta = \frac{\pi}{2}$$

$$= \frac{3}{2} \int_0^5 z^2 \times \frac{1}{2} \left[\frac{\frac{3}{2} \cdot \frac{1}{2}}{\frac{1}{2}} \right] dz$$

$$= \frac{3}{4} \int_0^5 z^2 \times \frac{1}{2} \pi dz$$

$$= \frac{3\pi}{8} \int_0^5 z^2 dz$$

$$= \frac{125\pi}{8} //$$

OP,

$$\text{Stokes, } \vec{F} = y^3 \vec{i} + x^3 \vec{j} + z^3 \vec{k}, C: (1,0,0), (0,1,0), (0,0,1)$$

$$(\nabla \times \vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^3 & 0 & x^3 \end{vmatrix} = 0 \cdot \vec{i} - 3x^2 \vec{j} - 3y^2 \vec{k}$$

$$\vec{r} = x\vec{i} + y\vec{j} + (1-x-y)\vec{k}$$

$$\vec{r}_x = \vec{i} + 0\vec{j} - \vec{k}, \quad \vec{r}_y = 0\vec{i} + \vec{j} - \vec{k}$$

$$\vec{N} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix}$$

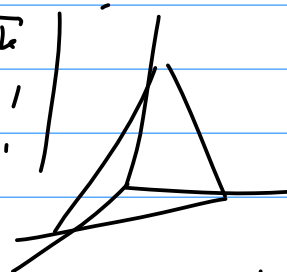
$$= \vec{i} + \vec{j} + \vec{k}$$

So, Take projection on xy, z=0

$$y = 1-x$$

$$x = 1$$

$$\therefore x + y + z = 1$$



$$(\nabla \times \vec{F}) \cdot \vec{N} = -3x^2 - 3y^2$$

$$\int_0^1 \int_0^{1-x} -3(x^2 + y^2) dy dx = -3 \int_0^1 \left(x^2 y + \frac{y^3}{3} \right) \Big|_0^{1-x} dx$$

$$= -3 \int_0^1 \left(x^2(1-x) + \frac{(1-x)^3}{3} \right) dx$$

$$= -3 \int_0^1 \left(x^2 - x^3 + \frac{1}{3}(1 - 3x + 3x^2 - x^3) \right) dx$$

$$= -3 \int_0^1 \left(2x^2 - \frac{4x^3}{3} - x + \frac{1}{3} \right) dx$$

$$= -3 \left[\frac{2x^3}{3} - \frac{x^4}{3} - \frac{x^2}{2} + \frac{x}{3} \right]_0^1$$

$$= -\frac{1}{2} //$$

6.b) find Fourier cosine series of $f(x) = x \sin x$, $0 < x < \pi$ → even
 Show $\frac{\pi \cdot 2}{4} = \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \frac{1}{7 \cdot 9} + \dots$

Soln,

The Fourier cosine series is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x \sin x dx$$

$$= \frac{1}{\pi} \left[-x \cos x + \sin x \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\pi \right] = 1$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \sin x \cos nx dx \Rightarrow \frac{1}{\pi} \int_0^{\pi} x \cdot 2(\cos nx \sin x) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \left[\sin(n x + x) - \sin(n x - x) \right] dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin(n+1)x dx - \frac{1}{\pi} \int_0^{\pi} x \sin(n-1)x dx$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[-x \frac{\cos(n+1)x}{(n+1)} + \frac{\sin(n+1)x}{(n+1)^2} - \left(-x \frac{\cos(n-1)x}{(n-1)} + \frac{\sin(n-1)x}{(n-1)^2} \right) \right]_0^\pi \\
&= \frac{1}{\pi} \left[-x \frac{\cos(n+1)x}{(n+1)} + \frac{\sin(n+1)x}{(n+1)^2} + x \frac{\cos(n-1)x}{(n-1)} - \frac{\sin(n-1)x}{(n-1)^2} \right]_0^\pi \\
&= \frac{1}{\pi} \left[\frac{-\pi(-1)^{n+1}}{(n+1)} + 0 + \frac{\pi(-1)^{n-1}}{(n-1)} - 0 - (0 + 0 + 0 - 0) \right] \\
&= \frac{(-1)^{n-1}}{(n-1)} - \frac{(-1)^{n+1}}{(n+1)} \\
&= -\frac{(-1)^n}{(n-1)} + \frac{(-1)^n}{(n+1)} \\
&= \frac{(-1)^n (n-1 - n-1)}{n^2-1} = \frac{-2}{n^2-1} (-1)^n = \frac{2}{n^2-1} (-1)^{n+1} \\
&\quad \text{for } n \geq 2
\end{aligned}$$

For $n=1$

$$\begin{aligned}
a_1 &= \frac{2}{\pi} \int_0^\pi x \sin x \cos x dx = \frac{1}{\pi} \int_0^\pi x \sin 2x dx \\
&\Rightarrow \frac{1}{\pi} \left[-x \frac{\cos 2x}{2} + \frac{\sin 2x}{4} \right]_0^\pi \\
&= \frac{1}{\pi} \left[-\frac{\pi}{2} - (0) \right] \\
&= -\frac{1}{2}
\end{aligned}$$

So,

$$f(x) = 1 - \frac{1}{2} \cos x + \sum_{n=2}^{\infty} \left[\frac{2(-1)^{n+1}}{n^2-1} \cos nx \right]$$

$$x \sin x = 1 - \frac{1}{2} \cos x + 2 \left[-\frac{\cos 2x}{3} + \frac{\cos 3x}{8} - \frac{\cos 4x}{15} + \frac{\cos 5x}{24} - \frac{\cos 6x}{35} + \dots \right]$$

Put $x = \frac{\pi}{2}$

$$\frac{\pi}{2} = 1 - 0 + 2 \left[\frac{1}{1 \cdot 3} + 0 - \frac{1}{3 \cdot 5} + 0 + \frac{1}{5 \cdot 7} - \dots \right]$$

$$\frac{\pi}{2} - 1 = 2 \left(\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots \right)$$

$$\therefore \frac{\pi - 2}{4} = \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots //$$

7.1) angle betⁿ surfaces = ? $a(1, -2, 1)$

$$\phi_1 = xy^2z - 3x - z^2 \quad , \quad \phi_2 = 3x^2 + 2z - 1 - y^2$$

$$\nabla\phi_1 = (y^2z - 3)\vec{i} + 2xy^2z\vec{j} + (xy^2 - 2z)\vec{k} \quad , \quad \nabla\phi_2 = 6x\vec{i} - 2y\vec{j} + 2\vec{k}$$

$$\text{at } (1, -2, 1)$$
$$\nabla\phi_1 = \vec{i} - 4\vec{j} + 2\vec{k} \quad , \quad \nabla\phi_2 = 6\vec{i} + 4\vec{j} + 2\vec{k}$$

$$\cos \theta = \frac{|\nabla\phi_1 \cdot \nabla\phi_2|}{|\nabla\phi_1| \cdot |\nabla\phi_2|} = \frac{(1, -4, 2) \cdot (6, 4, 2)}{\sqrt{1^2 + 4^2 + 2^2} \cdot \sqrt{6^2 + 4^2 + 2^2}} = \frac{6 - 16 + 4}{14\sqrt{6}} = \frac{-6}{14\sqrt{6}}$$

$$\theta = \cos^{-1}\left(\frac{-6}{14\sqrt{6}}\right)$$

$$\therefore \theta = \underline{\underline{79.92^\circ}}$$